

GLOBAL ARITHMETIC NONVANISHING OF THE ES ODD RECEIVING DETERMINANT

THE CLANKERS

ABSTRACT. For the literal prime–denominator quartic of an integral Erdős–Straus witness, we prove strict negativity of the original odd-receiver determinant numerator at every prime $p \equiv 1 \pmod{12}$. Explicit universal constants give $-\mathfrak{N} > cp^8$ and $\|O^{-1}\|_2 < 360S^9/(cp^8)$. The proof uses the complete Type I/II domain and two exact positive-coefficient polynomial certificates. This settles the arithmetic nonvanishing question isolated in the supplied programme audit, without its three-character restriction. It does not prove that an ES witness exists at every prescribed prime. Full integer coefficients, source maps, negative controls and two standard-library verification procedures are supplied.

1. THE ARITHMETIC QUESTION AND THE RESULT

The preceding receiver calculation isolates a polynomial F governing the original sign-odd receiving matrix of a quartic factor-inverse construction [2, EZ37–EZ45]. A later audit proves nonvanishing on a three-character prime locus and also constructs positive real and p -adic zeros outside the integral locus. We use its literal polynomial, roots, signs and coefficient normalization, read directly in the pinned public continuation [4, Track I, E1–E6]. The present result settles its integral nonvanishing question; it does not assert the existence of an Erdős–Straus (ES) witness at every prime.

Theorem 1.1 (Global arithmetic receiving determinant). *Let $p \equiv 1 \pmod{12}$ be prime and let x, y, z be positive integers satisfying $4/p = 1/x + 1/y + 1/z$. Retain the four literal roots $\ell = (p, x, y, z)$ and put*

$$S = p + x + y + z, \quad H(T) = -S^{-1} \prod_{i=0}^3 (T - \ell_i) = AT^4 + T^3 + BT^2 + CT + D.$$

Define F by (8) and $\mathfrak{N} = S^6 F$. Then $\mathfrak{N}$ is an integer and

$$\boxed{\mathfrak{N} < -c_{12}p^8 < 0, \quad c_{12} = \frac{125873811}{262144}.} \quad (1)$$

If $p \equiv 1 \pmod{24}$, one may replace c_{12} by

$$\boxed{c_{24} = \frac{327448292668}{47045881}.} \quad (2)$$

For the original odd receiving matrix O , every retained choice of root order and square-root signs gives

$$|\det O| > \frac{4cp^8}{S^3}, \quad \|O^{-1}\|_2 < \frac{360}{c} \frac{S^9}{p^8},$$

where c is the applicable constant. These constants are not asserted optimal on the original integral witness domain.

The proof uses two finite, universal polynomial coefficient identities. The complete integer arrays and two separate standard-library checks are supplied. The arithmetic census is a corroboration of the implementation, not the proof of the universal coefficient identities or the theorem.

2. ORIGINAL STATES AND THE COMPLETE FIRST-HALF DOMAIN

Here we give the elementary reductions needed to apply the theorem to every witness, rather than assume that its distinguished denominator has been chosen in a restricted family. The Type I/II framework is classical [1, §2, Propositions 2.2 and 2.6].

Every denominator exceeds $p/4$. At least one is divisible by p , by clearing the reciprocal equation; all three cannot be, since their reciprocal sum would then be at most $3/p$. A denominator equal to p would give

$$(3y - p)(3z - p) = p^2.$$

Here $y, z > p/3$, so both displayed factors are positive. The positive factors of p^2 are $1 \pmod{3}$, whereas each displayed factor is $2 \pmod{3}$. Hence no denominator equals p . If two denominators equal a , positivity of the third denominator gives $d = 2a - p > 0$. Then $4dz = p(d + p)$ and $d \mid p^2$. The three possibilities $d = 1, p, p^2$ give respectively $z = p(p + 1)/4, p/2, (p + 1)/4$, all nonintegral for $p \equiv 1 \pmod{4}$. Thus all four literal roots are distinct.

If two denominators are multiples of p , the third, denoted by a , must satisfy $a < p/2$: otherwise the sum is at most $4/p$, with equality requiring the impossible integer $a = p/2$. The other two denominators exceed a .

If exactly one denominator is pk , let the other two be $a \leq b$, both units at p . The identity

$$(4k - 1)ab = pk(a + b)$$

gives $4k - 1 = pt$, where $t \equiv 3 \pmod{4}$ and $\gcd(k, t) = 1$. Write $a = gr, b = gs, r \leq s$, $\gcd(r, s) = 1$. Then $tgrs = k(r + s)$, so $k = drs, r + s = t\lambda$ and $g = d\lambda$: indeed $\gcd(rs, r + s) = 1$ gives $rs \mid k$, say $k = drs$; then $tg = d(r + s)$, and $\gcd(d, t) = 1$ gives the remaining two divisibilities. If $a \geq p/2$, substitution of $p = (4drs - 1)/t$ and $a = d\lambda r$ gives

$$2dr(r - s) \geq -1.$$

The left side is an even nonpositive integer, forcing $r = s = 1$ and $t \mid 2$, a contradiction. Thus again $p/4 < a < p/2$, and a is the unique smallest original denominator.

Retain this a and $R = 4a - p$, so $3 \leq R < p$, $R \equiv 3 \pmod{4}$ and $\gcd(pa, R) = 1$. Since both tails exceed the unique smallest denominator, the two factor coordinates

$$X = Ry - pa, \text{quad} Z = Rz - pa$$

are positive and satisfy

$$XZ = p^2 a^2.$$

Because $p \nmid R$, one has $p \mid X$ exactly when $p \mid y$, and likewise for Z, z . Their p -valuation pairs are therefore $(0, 2), (1, 1), (2, 0)$. Place the unique p -divisible exterior denominator last. In the exterior and oriented middle cases, respectively, there is a unique positive divisor $u \mid a^2$ for which

$$E : (X, Z) = (a^2/u, p^2 u), \text{quad} M : (X, Z) = (pa^2/u, pu).$$

Solving for the tails gives

$$E : (y, z) = \left(\frac{pa + a^2/u}{R}, \frac{pa + p^2 u}{R} \right), \quad M : (y, z) = \left(\frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

Since a, u, p are units modulo R and $p \equiv 4a \pmod{R}$, the two exact integrality gates are

$$E : R \mid 4u + 1, \quad M : R \mid a + u. \tag{3}$$

Put

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g).$$

Primewise valuation gives $a = hrs, u = hr^2$, with all coordinates positive integers and $\gcd(r, s) = 1$. The ordered returns are

$$E : (a, hsk, phr\kappa), \quad \kappa = (pr + s)/R, \tag{4}$$

$$M : (a, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R. \tag{5}$$

Their ordered inverses are $u = a^2/(Ry - pa)$ and $u = pa^2/(Ry - pa)$, respectively. The middle complement $u \mapsto a^2/u$ exchanges r, s and only the last two denominators. Both orientations

remain recorded. No prime-factor exponents are identified across this map. The exterior case and the two middle orientations exhaust every positive integral witness.

For E, the same rational identity exists even before the full gate:

$$(a, y, z) = \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right). \quad (6)$$

Every original E word has $u \geq 2$, since $u = 1$ would force $R \mid 5$. Moreover, write $(\cdot/R)$ for the Jacobi symbol and $(\cdot/p)$ for the Legendre symbol. Then

$$\left(\frac{u}{p} \right) = -1. \quad (7)$$

For an odd prime $q \mid a$, reciprocity and $R \equiv -p \pmod{q}$ give

$$\left(\frac{q}{R} \right) = (-1)^{(q-1)/2} \left(\frac{R}{q} \right) = (-1)^{(q-1)/2} \left(\frac{-p}{q} \right) = \left(\frac{p}{q} \right) = \left(\frac{q}{p} \right).$$

If $2 \mid a$, the supplementary law and $R \equiv -p \pmod{8}$ give the same identity at two. Multiplication yields $(u/R) = (u/p)$, while the E gate gives $u \equiv -4^{-1} \pmod{R}$. Hence $(u/R) = (-1/R) = -1$. This proves (7). When $p \equiv 1 \pmod{24}$, two and three are residues, so $u \geq 5$. This character use is elementary and independent of the three receiving-frame characters in the audit.

3. THE LITERAL RECEIVER AND ITS INTEGER DETERMINANT

Let e_i denote the elementary symmetric functions of the four literal roots. Then

$$A = -1/S, \quad B = -e_2/S, \quad C = e_3/S, \quad D = -e_4/S.$$

For $d_j = H'(\ell_j)$ choose and retain each sign $\xi_j \in \mathbb{C}^\times$ with $\xi_j^2 d_j = 1$. The source's odd column is

$$o_j = \xi_j(1, -id_j, Ad_j^2 + 2\ell_j d_j, i(7\ell_j^2 d_j - 13d_j^2))^T.$$

The supplied polynomial is

$$\begin{aligned} F = & 20(3C - B^2) + A(87B^3 - 282BC - 51D) \\ & + A^2(-28B^4 + 22B^2C + 431C^2 + 204BD) \\ & + A^3(224B^2D - 168BC^2 - 856CD) - 448A^4D^2. \end{aligned} \quad (8)$$

Define the signed product

$$\epsilon = A^2 \prod_{i < j} (\ell_j - \ell_i) \prod_i \xi_i \in \{1, -1\}.$$

Then, exactly,

$$\det O = \frac{4\epsilon F}{A^3} = -\frac{4\epsilon \mathfrak{N}}{S^3}. \quad (9)$$

For completeness, reduce $1, H', A(H')^2 + 2TH', 7T^2H' - 13(H')^2$ modulo H in $1, T, T^2, T^3$. With the four row phases retained, the exact coefficient matrix is $\mathbf{C}_H = \text{diag}(1, -i, 1, i)\mathbf{M}_H \in M_4(\mathbb{C})$, where

$$\mathbf{M}_H = \begin{pmatrix} 1 & 0 & 0 & 0 \\ C & 2B & 3 & 4A \\ AC^2 - 9D & 4ABC - 8AD - 7C & 4AB^2 - 16A^2D - 2AC - 5B & 4AB - 8A^2C - 3 \\ 20D/A - 13C^2 & 20C/A + 76D - 52BC & 20B/A + 5C - 52B^2 + 208AD & 20/A - 66B + 104AC \end{pmatrix}.$$

Direct expansion gives $\det \mathbf{M}_H = 4F/A$; the phases $-i, i$ have product one. Evaluation at the roots and multiplication by the retained ξ_j gives (9), since

$$\prod_j d_j = A^4 \prod_{i < j} (\ell_j - \ell_i)^2.$$

The main verifier expands this complete Laurent-polynomial identity in A, B, C, D , rather than testing selected targets.

The integer numerator, used without assuming $p \nmid S$, is

$$\begin{aligned} \mathfrak{N} = & 60e_3S^5 - 20e_2^2S^4 + 87e_2^3S^2 - 282e_2e_3S^3 - 51e_4S^4 \\ & - 28e_2^4 + 22e_2^2e_3S + 431e_3^2S^2 + 204e_2e_4S^2 \\ & + 224e_2^2e_4 - 168e_2e_3^2 - 856e_3e_4S - 448e_4^2. \end{aligned} \quad (10)$$

It is homogeneous of degree eight in the literal roots. Root permutation does not change $\mathfrak{N}$, but the sign in (9) is retained.

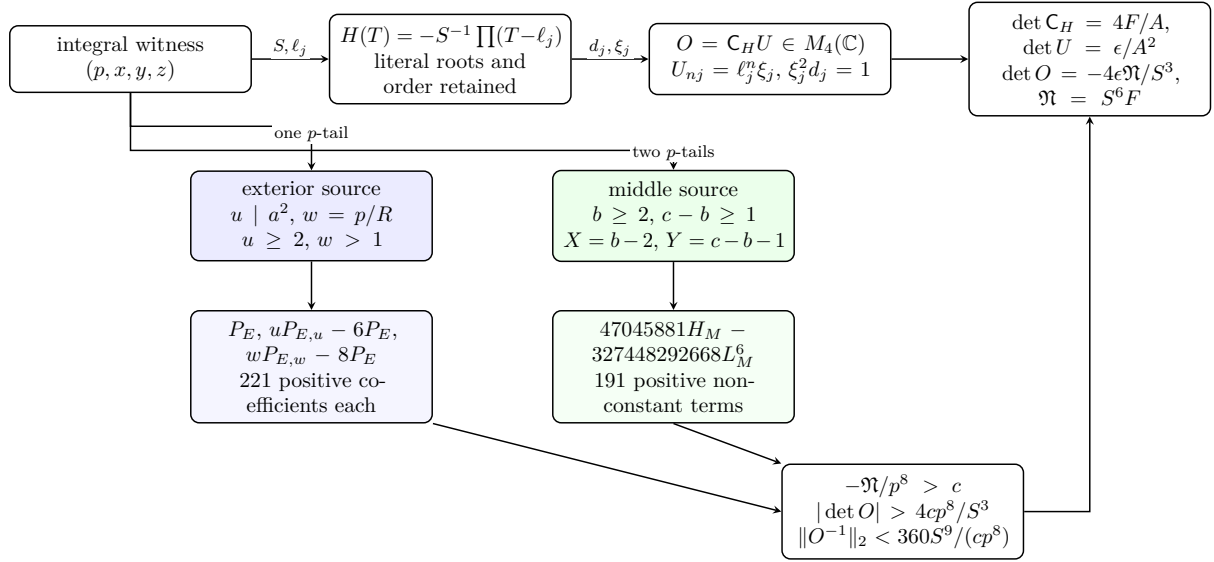


FIGURE 1. Typed route from a positive integral witness to the receiver bound. The two arithmetic branches exhaust the witness domain. The root labels, square-root signs, complex coefficient map and determinant phase are retained; the polynomial certificates control only the final real scalar $-\mathfrak{N}/p^8$.

4. EXTERIOR ARITHMETIC HAS A MONOTONE POSITIVE CERTIFICATE

In (6) set $w = p/R > 1$. Dividing the four roots by p gives

$$\left(1, \frac{w+1}{4w}, \frac{w+1}{4} + \frac{(w+1)^2}{16wu}, \frac{w+1}{4} + wu\right).$$

Define the exact polynomial P_E by

$$\begin{aligned} -64u^2 P_E(u, w) = & \mathfrak{N}(16wu, 4u(w+1), \\ & (w+1)(4wu + w + 1), 4wu(w+1 + 4wu)). \end{aligned} \quad (11)$$

The right side is divisible by $64u^2$ in $\mathbb{Z}[u, w]$; the sign on the left is part of the definition. The resulting bidegree is $(12, 16)$. Hence

$$-\frac{\mathfrak{N}(p, a, y, z)}{p^8} = Q_E(u, w) := \frac{P_E(u, w)}{2^{26}u^6w^8}. \quad (12)$$

Lemma 4.1 (Finite universal exterior coefficient certificate). *Each of*

$$P_E(1 + X, 1 + Y), \quad (u\partial_u P_E - 6P_E)(1 + X, 1 + Y), \quad (w\partial_w P_E - 8P_E)(1 + X, 1 + Y)$$

has exactly 221 coefficients, one at every position $0 \leq i \leq 12$, $0 \leq j \leq 16$, and every coefficient is a positive integer.

Exact certificate and its independent verification. Equation (10) and the four integer polynomials in (11) define every coefficient by finite integer additions and multiplications. The complete arrays are in `polynomial_certificates.json`. The first verifier performs the division and binomial translation exactly. The second evaluates the unexpanded S^6F on the entire 17×17 grid $u, w \in \{1, \dots, 17\}$. Both sides of (11) have bidegree at most $(16, 16)$ before division. Univariate polynomial uniqueness applied successively in the two variables proves equality from that complete grid. It then independently performs both derivative operations and all binomial translations. Table 1 displays the row minima; no untested sign or floating-point comparison is used. This is a finite computer-assisted polynomial lemma, not an extrapolation from the prime census. $\square$

The derivative numerators in Lemma 4.1 show that Q_E strictly increases in both u and w on $[1, \infty)^2$. Its exact boundary value is

$$\Phi(u) = Q_E(u, 1) = \frac{(2u-1)^2}{4096u^6} V(u), \quad (13)$$

where

$$\begin{aligned} V(u) = & 5120u^{10} + 14592u^9 + 7232u^8 - 6400u^7 + 1728u^6 \\ & + 3904u^5 + 432u^4 - 400u^3 + 113u^2 + 57u + 5. \end{aligned}$$

The boundary equality is also a coefficient identity, independently checked at 13 arguments after its degree-12 polynomial denominator is cleared. In particular,

$$\Phi(2) = c_{12}, \quad \Phi(5) = \frac{1310003654463}{12800000} > c_{24} > c_{12}.$$

Every actual E word has $u \geq 2$ and $w > 1$, so (1) follows for E. At $p \equiv 1 \pmod{24}$, use $u \geq 5$ for the stronger constant. More generally (12) proves nonvanishing throughout the unfiltered rational E source with real $u \geq 1, w \geq 1$, even when its original integer gate fails.

5. MIDDLE ARITHMETIC HAS A UNIFORM SEPARATION CERTIFICATE

For M, reorder only the last two denominators and write them as pb, pc with integers $b < c$. The original prime cannot be a denominator, so $b \geq 2$ and $c - b \geq 1$. The normalized smallest denominator is

$$a/p = bc/L, \quad L = 4bc - b - c > 0.$$

For normalized denominator sum $s = (a + pb + pc)/p$ and product $v = a(pb)(pc)/p^3$, set

$$T = (b + c)L + bc, \quad V = b^2c^2, \quad s = T/L, \quad v = V/L.$$

Substitution into (10), using the exact normalized reciprocal identity, gives $-\mathfrak{N}/p^8 = \mathcal{P}(s, v)$, where

$$\begin{aligned} \mathcal{P}(s, v) = & 7168v^4 - (5568s^2 + 5728s - 5888)v^3 \\ & + (320s^4 + 2744s^3 + 705s^2 - 3350s - 903)v^2 \\ & + (-140s^5 - 443s^4 + 440s^3 + 390s^2 + 70s - 249)v \\ & + 20s^6 - 7s^5 - 26s^4 - 7s^3 + 20s^2. \end{aligned} \quad (14)$$

Define the integer polynomial

$$H(b, c) = L^6 \mathcal{P}(T/L, V/L), \quad H_M(X, Y) = H(2 + X, 3 + X + Y),$$

and retain the denominator

$$L_M = 4X^2 + 4XY + 18X + 7Y + 19.$$

Lemma 5.1 (Finite universal middle coefficient certificate). *H_M has 192 nonzero positive integer coefficients. The polynomial*

$$47045881H_M - 327448292668L_M^6 \quad (15)$$

has zero constant coefficient and exactly 191 nonzero coefficients, all positive integers. In particular it is nonnegative on $X, Y \geq 0$, and strictly positive unless $X = Y = 0$.

Exact certificate and its independent verification. Use (14) to generate the first array by integer polynomial operations. The second is its literal difference with the sixth power of the displayed L_M . Table 2 records every row's nonzero count and minimum. The complete arrays include the positive coefficients of X and Y . Independently, with $b = 2 + X, c = 3 + X + Y$, evaluate

$$\mathfrak{N}(L, bc, bL, cL) = -L^2 H_M(X, Y)$$

on the full grid $0 \leq X \leq 24, 0 \leq Y \leq 16$. Each scaled root has degree at most three in X and two in Y ; (10) is homogeneous of degree eight. The left side thus has bidegree at most $(24, 16)$; the array bounds give the same bound on the right. The complete grid therefore proves the polynomial identity. The independent checker multiplies L_M^6 separately and confirms all coefficients of (15). $\square$

Since $L_M > 0$, the lemma proves $-\mathfrak{N}/p^8 \geq c_{24}$ throughout the actual M domain. Equality in the larger real cone requires $b = 2, c = 3$, whence $a = 6p/19$. It would force $p = 19$ for prime p , outside 1 (mod 12). Thus the original inequality is strict. Combining this with the E result proves the determinant assertion of Theorem 1.1.

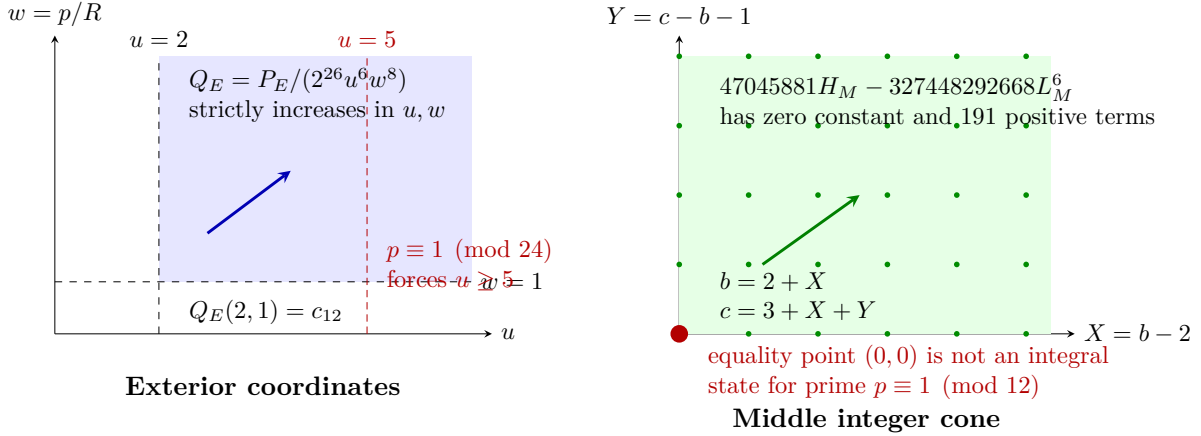


FIGURE 2. The two exact coordinate domains used in the global sign proof. The blue and green regions are certificate domains; actual Erdős–Straus witnesses are the arithmetic points satisfying the displayed gates and integrality conditions. The exterior lower bound is approached at the dashed boundary $w = 1$, which is excluded by $R < p$. The middle comparison is strict at every admissible lattice point because its only equality point would require $a = 6p/19$.

6. THE ORIGINAL INVERSE, SIGNS, NORMS, AND CONDUCTOR COMPOSITION

Let C_H be the displayed coefficient matrix of the four original component polynomials and put

$$U_{nj} = \ell_j^n \xi_j, \quad 0 \leq n, j \leq 3.$$

Then $O, C_H, U \in M_4(\mathbb{C})$ and

$$O = C_H U, \quad \det C_H = 4F/A, \quad \det U = \epsilon/A^2.$$

For given $z = O\beta$, compute $m = C_H^{-1}z$ by its actual adjugate and determinant. The complete signed inverse is

$$\begin{aligned} \beta_j = \xi_j [& (C + B\ell_j + \ell_j^2 + A\ell_j^3)m_0 + (B + \ell_j + A\ell_j^2)m_1 \\ & + (1 + A\ell_j)m_2 + Am_3]. \end{aligned} \quad (16)$$

This is Lagrange interpolation; $1/(\xi_j d_j) = \xi_j$. Its source comprises the literal root labels and their retained signs. Neither squared norms nor an unordered root list is asserted to reconstruct that source.

Here and in Theorem 1.1, $\|\cdot\|_2$ is the operator norm for the standard Hermitian norm on $\mathbb{C}^4$. For four distinct positive integer literal roots,

$$2/S \leq |d_j| \leq S^2.$$

Indeed $|d_j| = S^{-1} \prod_{k \neq j} |\ell_j - \ell_k|$. The three nonzero integer differences cannot all have magnitude one, while each is strictly less than S . The four row-entry bounds of O are therefore

$$\sqrt{S/2}, \quad S, \quad 3S^2, \quad 20S^3.$$

Every 3×3 cofactor has magnitude at most $360S^6$. The Frobenius norm of the sixteen-cofactor adjugate is at most $1440S^6$. Dividing by the determinant lower bound proves the inverse bound in Theorem 1.1. For original positive label and receiving metrics G and Q , respectively,

$$\|O^{-1}\|_{Q \rightarrow G} < \|G^{1/2}\|_2 \|Q^{-1/2}\|_2 \frac{360S^9}{cp^8}. \quad (17)$$

No new invariant metric is substituted. If a further original linear observation Λ is retained, then

$$\ker(\Lambda O) = O^{-1} \ker \Lambda, \quad \text{rank}(\Lambda O) = \text{rank} \Lambda.$$

The receiver adds no further kernel on this arithmetic domain; it does not assert that the observation itself is injective. On the image, for original label metric G , its quotient metric is exactly $(\Lambda O G^{-1} O^* \Lambda^*)^{-1}$. Transporting by O uses $O^{-*} G O^{-1}$ rather than assigning a new norm. For an original invertible conductor L , the complete inverse remains $(LO)^{-1} = O^{-1} L^{-1}$ with its actual second factor. An enlarged receiving map $\mathcal{J}(\alpha, \beta) = (\alpha, E\alpha + O\beta)$ likewise has the literal inverse $(\alpha, z) \mapsto (\alpha, O^{-1}(z - E\alpha))$.

There is also a bound depending only on the prescribed prime. Let

$$s_p = p(p+3)/4, \quad B_p = p + s_p(s_p + 1).$$

The tail factor identity bounds either tail by $pa(pa+1)/R$. Direct subtraction from its value at $R = 3$ gives

$$\frac{pa(pa+1)}{R} - \frac{s_p(s_p+1)}{3} = \frac{p^2(R-3)}{16} \left(1 - \frac{p^2+4}{3R}\right) \leq 0.$$

Since a is smallest, $S \leq B_p$. Thus (17) gives the explicit coefficient bound $360B_p^9/(cp^8) = O(p^{28})$. It is a bound on every existing integral witness, not an assertion that one exists. Homogeneous scaling gives $\mathfrak{N}_p = p^8 \mathfrak{N}_1$ and, with coherent signs, $O_p = \text{diag}(p^{-1}, p, p^2, p^3) O_1$; these scale factors are retained.

7. ARITHMETIC EXAMPLES AND PRECISE NONCLAIMS

At the hard prime $p = 1009$, the word $u = 11$ at $a = 253, R = 3$ gives

$$E : (253, 87032, 3818056), \quad M : (253, 2042216, 88792).$$

The audit's character triple is $(-1, 1, 1)$ at this prime. The present theorem nevertheless proves nonvanishing for every one of its integral witnesses. For these two states $v_p(\mathfrak{N})$ is respectively zero and two. At $p = 944329$, the original M state $(a, R, u) = (236094, 47, 1444)$ returns

$$(236094, 780326438241, 4772638766).$$

All three old characters are positive, while the new negative bound still holds. The exact integer numerators and original factor boxes appear in the certificate.

The theorem is not a claim that $\mathfrak{N}$ is a unit at p . In the complete census over primes $p \equiv 1 \pmod{12}$ through 3000, the valuation histogram is $E : v_p = 0$ for 2447 states, $E : v_p = 1$ for four states, and $M : v_p = 2$ for 2054 states. The arithmetic order/conductor defects studied earlier remain; nonvanishing over $\mathbb{Q}$ is a different assertion.

A positive real counterdomain also remains. For $3 \leq t \leq 4$, put

$$p = 1, \quad b = t, \quad c = t + 1/10, \quad a = bc/(4bc - b - c).$$

These are distinct positive reciprocal roots with $1/4 < a < 1/2 < 1 < t < t + 1/10$. Put $D = 40t^2 - 16t - 1$. Exact elimination gives

$$\mathfrak{N}(t) = \frac{P(t)}{100000D^6},$$

where, in descending order from t^{18} to the constant term, the coefficient vector of P is

$$\begin{aligned} &(3617792000000000, -26154803200000000, 70130040320000000, \\ &-88735357952000000, 45725157043200000, 12437702609920000, \\ &-28537312234432000, 13617485725683200, -1729738000411520, \\ &-478292746960768, 90264805813440, 17381353732960, -951549897144, \\ &-455951154664, -51247817566, -3092472120, -110106797, -2195730, -19035). \end{aligned}$$

All eighteen coefficients of $P'(3 + X)$ are positive. Exact endpoint signs therefore give a unique receiver singularity

$$\frac{3053}{1000} < t_0 < \frac{1527}{500}, \quad t_0 \approx 3.05367681075890,$$

while the quartic remains squarefree. The M separation $c - b \geq 1$ fails. The E raw word is also below one at both endpoints. Scaling by any prime $p > 5$ leaves the tail gap $p/10$ nonintegral, so these points cannot be promoted to integral witnesses. The audit's separate positive-real example and p -adic zeros are not contradicted by Theorem 1.1.

Finally, even an available divisor need not satisfy a gate. At $p = 1009, a = 321, R = 275, u = 9$, the raw exterior identity is

$$(321, 335338/275, 9486618/275).$$

Its rational quantity S^6F is strictly negative by Lemma 4.1, but $275 \nmid 37$: the E divisibility gate fails. Frame invertibility therefore does not force an original integer coefficient. The divisor-descent handoff's universal occupancy obligation is unchanged [3].

8. CERTIFICATE SCOPE AND REPRODUCIBILITY

Run the two separate implementations with the stated bounds:

```
python verify.py --bound 3000 --out certificates
python -O verify.py --bound 3000 --out certificates_optimized
python check_independent.py --input certificates \
  --out certificates/independent.json
```

The first program performs the universal Laurent-polynomial reduction, all coefficient identities and the original divisor scan. The second uses the unexpanded supplied F , complete unisolvent grids, separate binomial arithmetic, and a primitive (h, r, s) enumeration. It imports neither the first program nor predecessor software. Both use only the Python standard library and explicit checks that remain present under optimized execution.

The bounded arithmetic lane covers every first-half source at primes $p \equiv 1 \pmod{12}$ through 3000: 99 primes, 34287 shells, 868359 original divisor words, 2451 E states and 2054 oriented M states. Source hashes, execution outcomes and negative-control values are recorded separately. This is not a new overall ES verification range, an independent mathematical review, or a Lean build. The universal theorem is the proved arithmetic reduction together with the complete polynomial certificates.

The precise newly closed question is nonvanishing of the supplied odd receiving determinant at every positive integral hard-prime witness. The initial existence of such a witness is not a premise that can be discharged by this nonvanishing theorem. The work remains at the programme's Turn 7.

APPENDIX A. COMPACT COEFFICIENT LEDGER

The full arrays are machine-readable; the following minima make their signs visible without replacing the identity checks.

TABLE 1. Exterior coefficient minima. Each row contains 17 coefficients in each of the three arrays.

X degree	$P_E(1 + X, 1 + Y)$	$uP_{E,u} - 6P_E$ shifted	$wP_{E,w} - 8P_E$ shifted
0	414843750	1950703125	3318750000
1	4439765625	21170421875	35518125000
2	21684625000	105062450000	173477000000
3	63933650000	315267850000	511469200000
4	126767200000	637111680000	1014137600000
5	178129216000	913469734400	1425033728000
6	181933158400	952827043840	1455465267200
7	136118149120	728564637696	1088945192960
8	74055811072	405315584000	592446488576
9	28578217984	160000049152	228625743872
10	7426539520	42543874048	59412316160
11	1167065088	6841958400	9336520704
12	83886080	503316480	671088640

TABLE 2. Middle comparison polynomial: every nonzero coefficient is positive; its constant is zero.

Y degree	Nonzero count	Minimum coefficient
0	18	12224401719040
1	19	28134189572096
2	19	14067094786048
3	18	126603853074432
4	17	510269410869248
5	16	1212467950600192
6	15	1880365903044608
7	14	1988277863047168
8	13	1451415862034432
9	12	722239332302848
10	11	234515813076992
11	10	44899083358208
12	9	3853998571520

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